Solution Architecture

Date	06 July 2026
Team ID	
Project Name	Smart Lender - Loan Eligibility Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer

1.HTML

2.CSS

3.Bootstrap

4.JavaScript

5.Flask

[e.g., Web App (React), Mobile App (Flutter), Admin Dashboard]

↓↑

API Gateway / Load Balancer

1.Flask Application

2.REST API

[e.g., Nginx, AWS API Gateway, Express Server]

↓↑

Auth Service	Core Logic Service	External API Integrations
[e.g., JWT, OAuth2] *1.User login authentication* *2.Session management*	*[e.g., Data Processing, Machine Learning Models]* *1.Loan Eligibility Prediction* *2.Data preprocessing*	*[e.g., Payment Gateway, Email Service]* *1.Future : Email notification API*

2.Future : SMS notification API

↓↑

Data / Storage Layer

1.CSV Dataset

2.Loan Model.

3.SQLite

[e.g., PostgreSQL Database, MongoDB, AWS S3]

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	*Collects user loan details and displays prediction results*	*HTML,CSS, Bootstrap, javascript, Flask*
[API Gateway]	*Receives user requests and forwards them to the ML model*	*Flask ,REST API*
[Microservices]	*Performs data preprocessing*	*Python,Pandas Numpy,Scikitlearn, XGBoost*
[Database]	*Stores user details, datasets and trained model*	*SQLite and MySQL,CSV, Joblib*